

Controller design in the frequency domain

R0005E - Measurement and Feedback Control

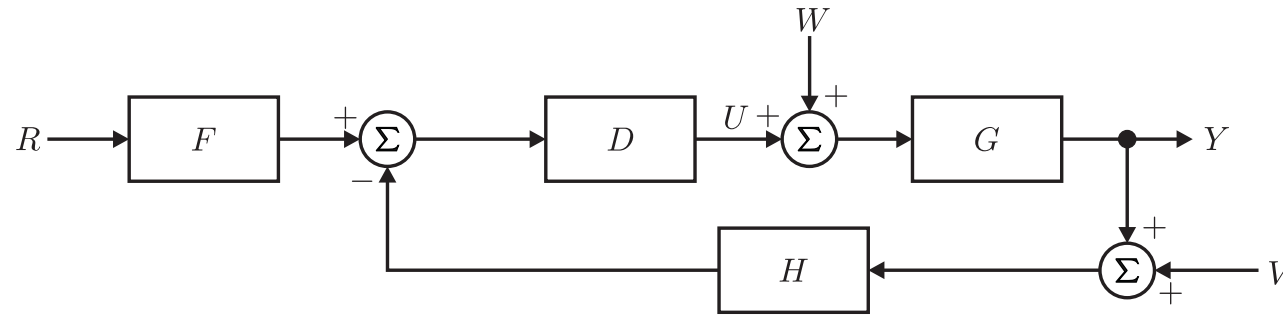
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Specification in the frequency domain

The closed loop system is given in the following general form:



Properties of a well designed closed-loop system:

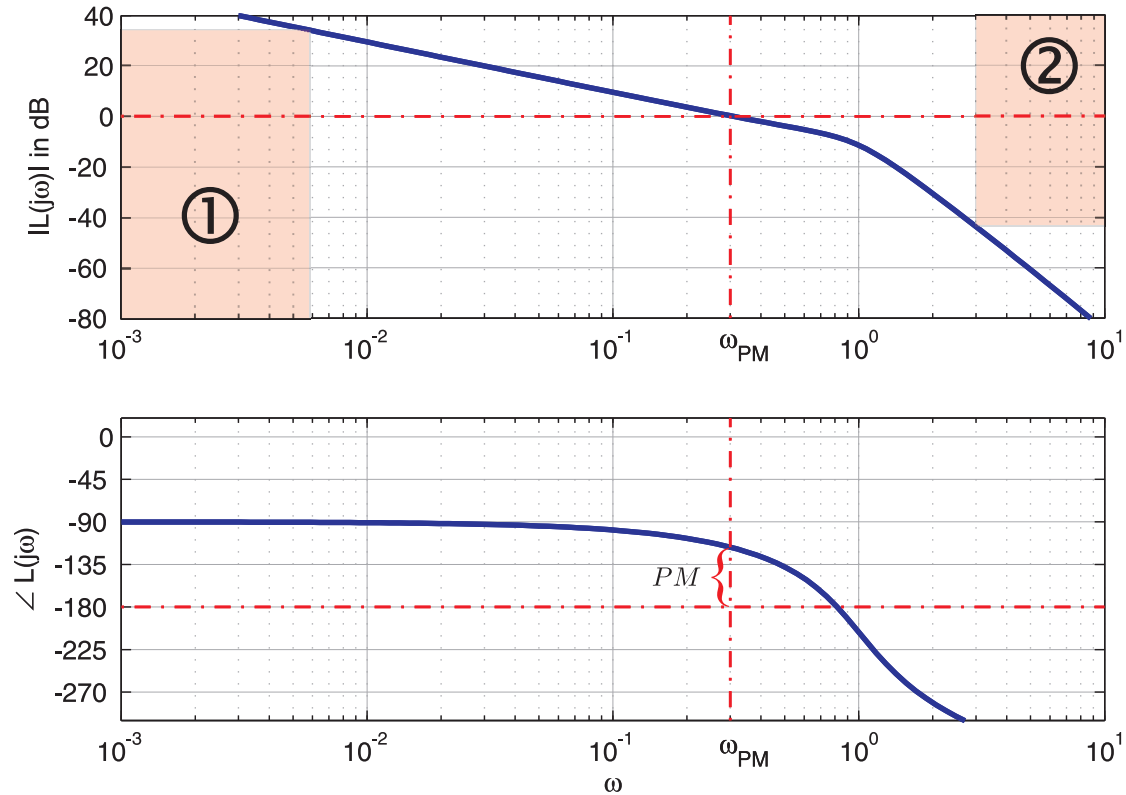
1. Stability
2. Small or zero steady state control error

These properties have a representation in the frequency domain. Moreover, they can be translated into properties of the open-loop system. Again, we define the open-loop transfer function as $L = GD$.

Stability $\triangleq$ L has a large phase margin (PM)

Small or zero steady state control error $\triangleq$ L has a large magnitude at zero frequency

In a Bode plot these two specifications are fulfilled:



Corresponding specifications:

1. **Stability:**

For the phase margin (PM) it is usual to require $PM \geq 45^\circ$. A large phase margin relates to a small or no peak in the closed loop transfer functions (Remember the vector margin!)

2. **Small or zero steady state control error:**

An error boundary can be defined, as depicted by region 1. The open-loop transfer function L must not enter that region.

3. **Insensitivity to noise and modeling errors:**

Additionally, one would like to reduce the sensitivity to sensor noise and to modeling errors. Region 2, could be understood as a boundary for these types of effects. The open-loop transfer function L must not enter that region. The design of such a specification will not be further discussed.

Note:

$$T(s) = \frac{Y(s)}{R(s)} = \frac{L(s)}{1 + L(s)}; \quad S(s) = \frac{E(s)}{R(s)} = \frac{1}{1 + L(s)}; \quad T(s) + S(s) = 1$$

Consequently, for a large $L(j\omega)$, $T(j\omega) \rightarrow 1$ and $S(j\omega) \rightarrow 0$.

Consequence: We can design a closed loop system, based on a specification on the open-loop transfer function L .

Important note: An often occurring error is to perform calculation on the closed loop transfer function!

Some calculation aspects

Working with Matlab:

- `bode` and `nyquist` can be used to plot the frequency response
- For manual plotting the function `semilogx` of the phase and the magnitude in dB, or `loglog` for plotting the magnitude in the absolute scale.
- `mag2db` can be used to perform the conversion into dB.
- `margin` calculates the phase and gain margin including the frequencies at which they occur.

Doing hand calculations:

- Use the asymptotic plot of the frequency response to get a rough understanding of the shape of the frequency response. Then you can estimate the frequency region for the cross-over quite well.
- The frequency response is usually a monotone function in the vicinity of the cross-over.
- Derive $|L(j\omega)|$ and $\angle L(j\omega)$ as explicit functions of ω
- Do some iterations to calculate ω_{PM} or ω_{GM} . Subsequently PM and GM can be derived.
- Dependencies on a gain K in a controller can usually be dealt with.

Controller design

Since controllers D are achieving a compensation of the open-loop transfer function, they are often called compensators.

Design for increased stability

- PD Controller
- Lead Compensator

Design for tracking and disturbance rejection

- PI Controller
- Lag Compensator

Increased stability, tracking and disturbance rejection

- PID Controller
- Filtered PID Controller

PD Controller

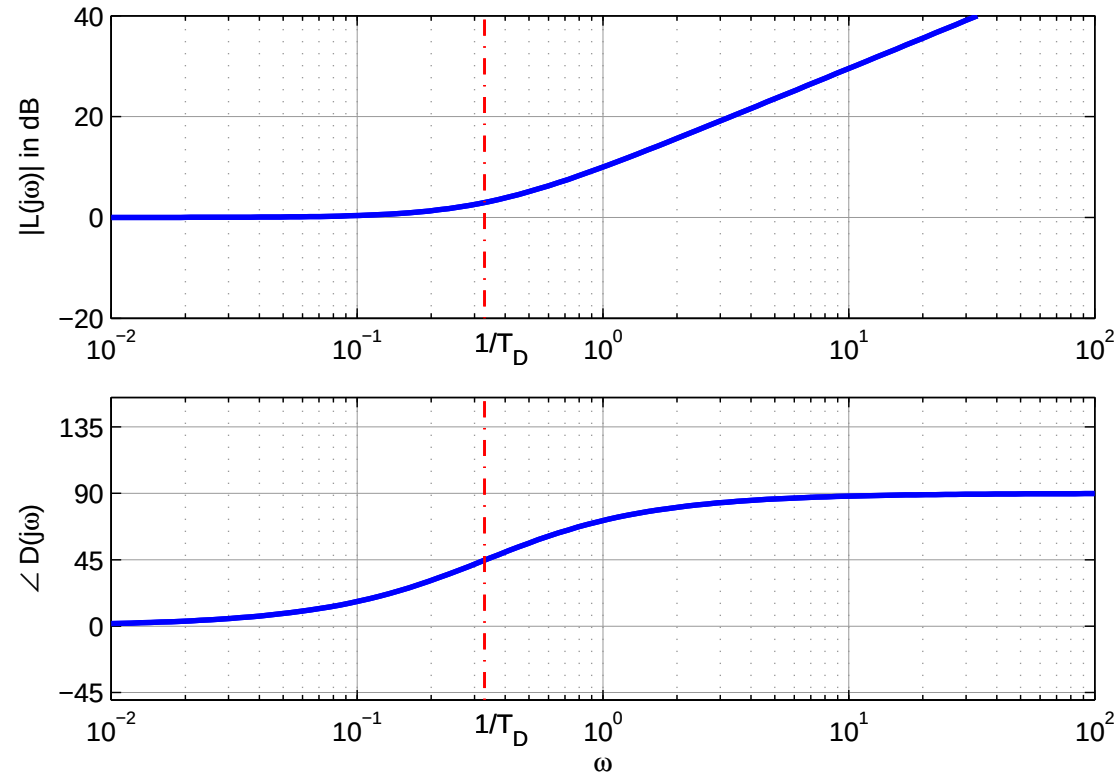
The PD Controller is given by

$$D(s) = K(1 + T_D s)$$

We can translate $D(s)$ into the Bode form and get

$$D(j\omega) = K(j\omega T_D + 1)$$

This is a type 2 term with a positive exponent:



Observations:

- The controller has a high gain at high frequencies
- It provides a positive phase contribution

Design Guideline:

- Choose T_D such that $\frac{1}{T_D}$ coincides with the cross-over frequency of L .
- Choose K such that the steady state control error condition is fulfilled, that is for $\omega = 0$.

Advantage:

- The PD increases the phase margin of the system and therefore improves stability.

Drawback:

- The PD amplifies high frequencies, which implies a sensitivity to sensor noise

Lead Compensator

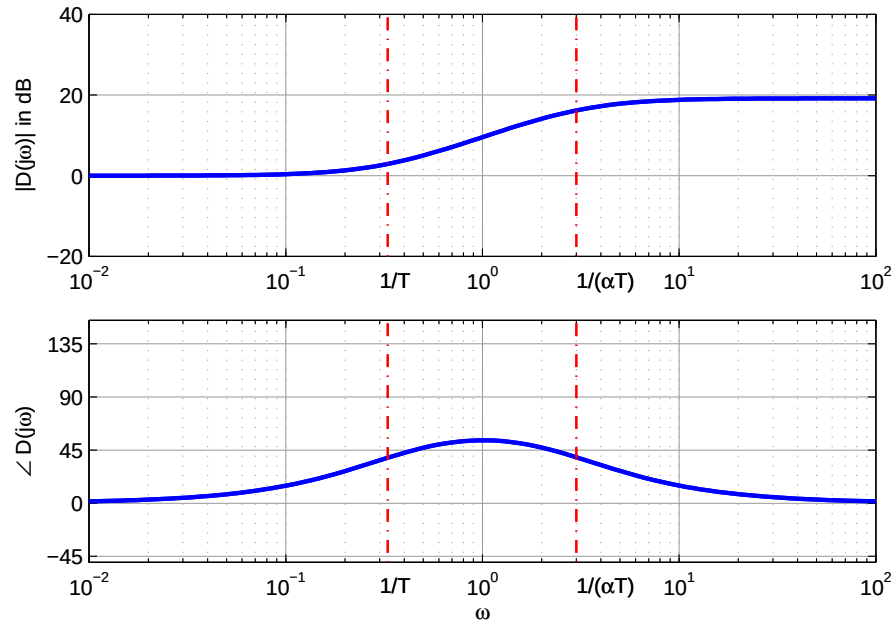
The Lead Compensator is given by

$$D(s) = K \frac{Ts + 1}{\alpha Ts + 1}; \quad \alpha < 1$$

We can translate $D(s)$ into the Bode form and get

$$D(j\omega) = K \frac{j\omega T + 1}{j\omega \alpha T + 1}$$

This is a combination of two type 2 terms with different break points and exponents:



Observations:

- The gain change is equal to $\frac{1}{\alpha}$.
- The controller increases for high frequencies in a rather limited way.
- It provides a positive phase contribution

$$\varphi = \angle D(j\omega) = \tan^{-1}(T\omega) - \tan^{-1}(\alpha T\omega)$$

which yields a maximum contribution of

$$\varphi_{max} = \sin^{-1} \frac{1 - \alpha}{1 + \alpha} \quad \text{at} \quad \omega_{max} = \frac{1}{T\sqrt{\alpha}}$$

Design Guideline:

- First choose K such that the steady state control error condition is fulfilled, that is for $\omega = 0$.
- Choose α such that an acceptable contribution to the phase margin is provided at the cross-over frequency of L .

Advantage:

- The Lead compensator increases the phase margin of the system and therefore improves stability.

Drawback:

- The Lead compensator amplifies high frequencies for phase contribution above 70° .
- The achieved phase margin increase is usually less than the expected value.

PI Controller

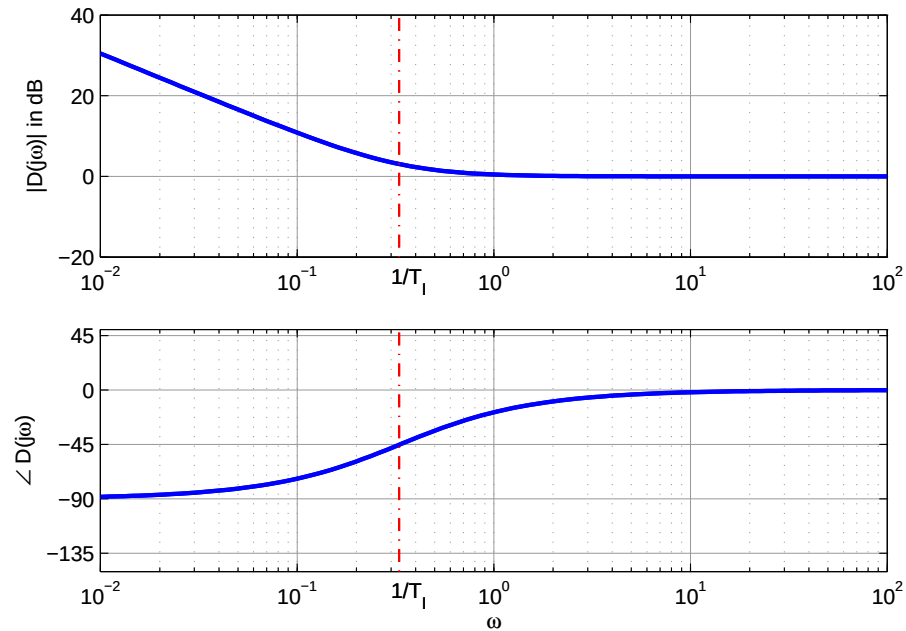
The controller is given by

$$D(s) = K\left(1 + \frac{1}{T_I s}\right) = K \frac{T_I s + 1}{T_I s}$$

The Bode of the controller is then

$$D(j\omega) = K \frac{j\omega T_I + 1}{j\omega T_I}$$

This is a combination of a type 1 and a type 2 term.



Observations:

- Gives a small steady state control errors, since the gain is infinite at zero frequency.
- The controller has a negative phase contribution

Design Guideline:

- First choose K to determine the cross-over of the system with a certain phase margin. A faster response requires a higher frequency for the cross-over.
- Choose T_I such that the cross-over frequency is substantially larger than $\frac{1}{T_I}$ and the negative phase contribution does not affect the phase margin too much.

Advantages:

- Zero steady state control error

Drawback:

- Negative phase contribution which usually leads to oscillations.

Lag Compensator

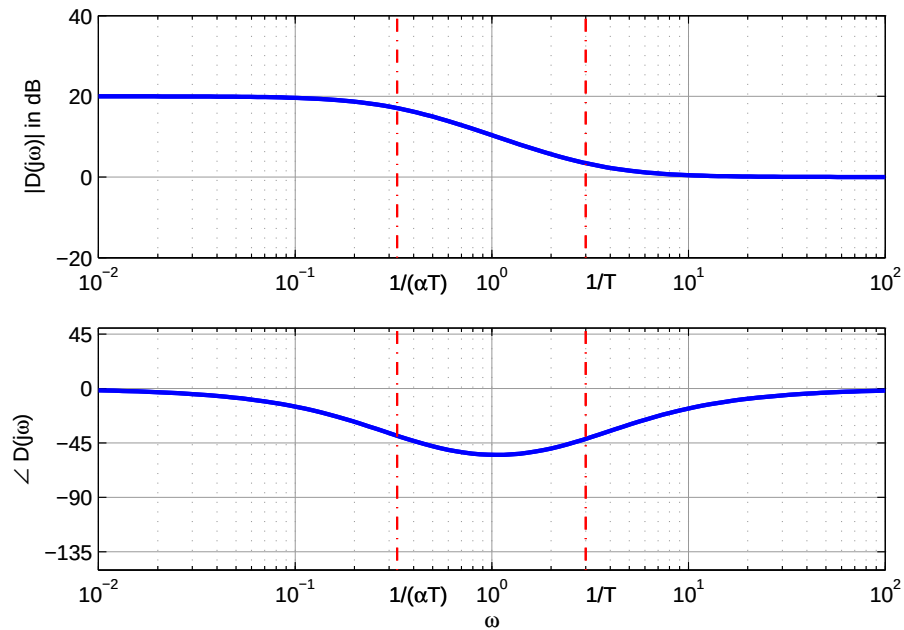
The compensator transfer function is given as

$$D(s) = K\alpha \frac{Ts + 1}{\alpha Ts + 1}; \quad \alpha > 1$$

and the resulting Bode form is

$$D(j\omega) = K\alpha \frac{j\omega T + 1}{j\omega \alpha T + 1}$$

Similar to the Lead compensator we have a combination of two type 2 terms



Observations:

- The compensator has a gain change of α , with a higher gain for low frequencies.
- The controller has a negative phase contribution

Design Guideline:

- First choose K to determine the cross-over of the system with a certain phase margin. A faster response requires a higher frequency for the cross-over.
- Choose α to meet the steady state control error requirements.
- Choose T such that the cross-over frequency is approximately one octave to one decade higher than $\frac{1}{T}$
- If the design does not fulfill the requirements, especially regarding the phase margin, you need to iterate and adjust the parameters

Advantages:

- Small steady state control error.
- Reduced negative phase contribution, compared to the PI controller.

Drawback:

- Complicated and iterative design procedure.

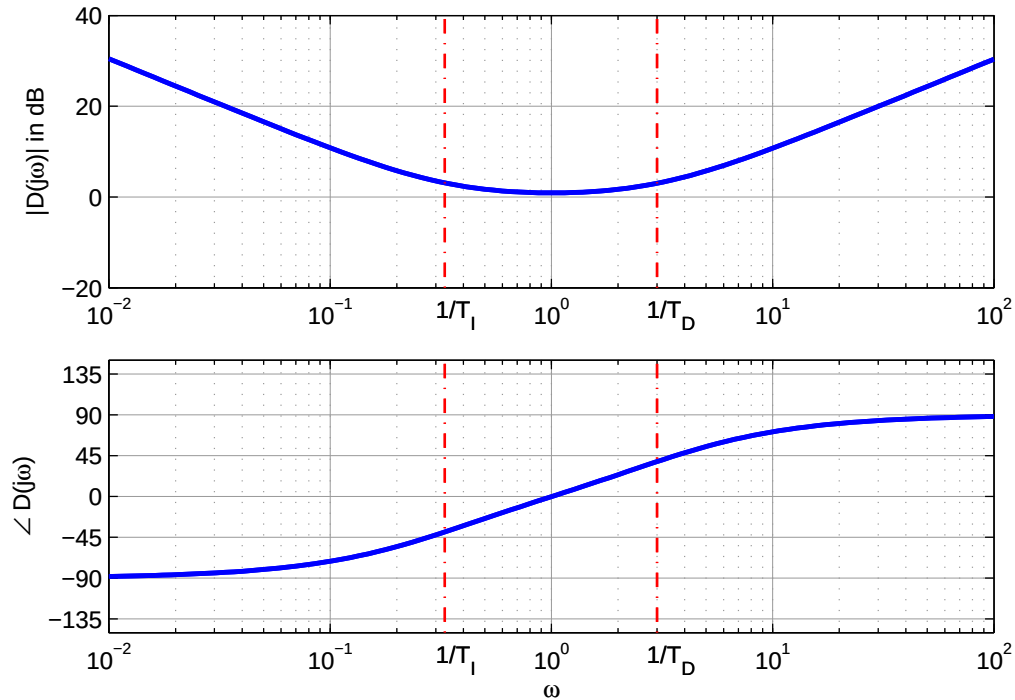
PID Controller

The PID controller could be given in the following form:

$$D(s) = K \left[(T_D s + 1) \left(1 + \frac{1}{T_I s} \right) \right]$$

which differs from the earlier introduced forms. But the parameters could be converted into the other forms as well.

The bode form is then a series of PD and PI controller.



Observations:

- The controller combines the affects of a PD and a PI controller
- The negative phase contribution of the PI can be reduced by the PD part.

Design Guideline:

- First choose K and T_D to determine the cross-over of the system with a certain phase margin. A faster response requires a higher frequency for the cross-over.
- Choose T_I such that the negative phase contribution does not affect the phase margin too much.

Advantages:

- Small or zero steady state control error.
- Usually a good phase margin can be achieved.

Drawback:

- Amplifies high frequencies and can be sensitive to sensor noise.

Filtered PID Controller

In industrial applications, sensor noise can be problematically high, which usually makes the PID controller too sensitive.

Ideas:

- Replace the PD part of the controller by a Lead compensator.
- Add a Low pass filter to the PID controller.

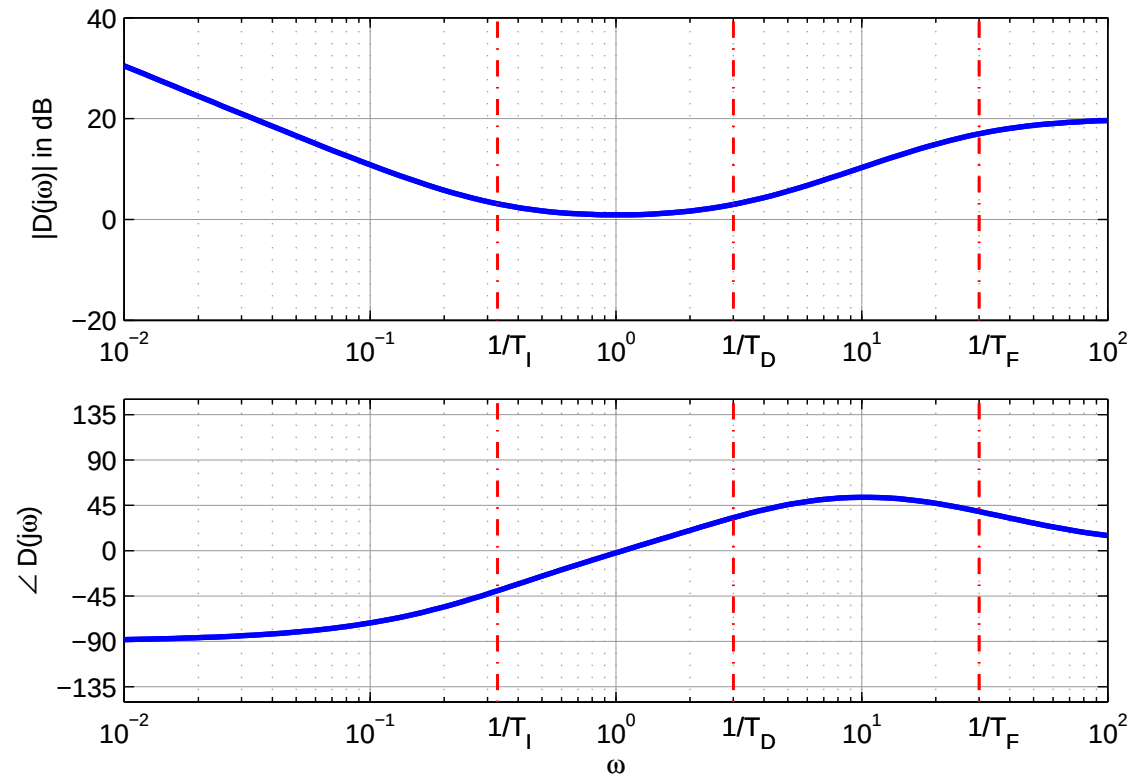
In case of adding a low pass filter one gets

$$D(s) = K \left[(T_D s + 1) \left(1 + \frac{1}{T_I s} \right) \frac{1}{T_f s + 1} \right]$$

which renders into the Bode form

$$D(j\omega) = K \left[(j\omega T_D + 1) \frac{j\omega T_I + 1}{j\omega T_I} \frac{1}{T_f s + 1} \right]$$

Clearly, the term $(j\omega T_D + 1)$ and $(T_f s + 1)^{-1}$ constitute a Lead compensator, since $T_f < T_D$ and could be written as $T_f = \alpha T_D$, with $\alpha < 1$.



Observations:

- The controller combines the affects of a PD and a PI controller
- Limits the noise sensitivity.

Design Guideline:

- First choose K and T_D to determine the cross-over of the system with a certain phase margin. Avoid high frequencies for the cross-over. Analyze the sensor noise prior to the design.
- Choose T_F such that the observed noise is attenuated and the cross-over is at least one octave below $\frac{1}{T_F}$.
- Choose T_I such that the negative phase contribution does not affect the phase margin too much.

Advantages:

- Small or zero steady state control error.
- Usually a good phase margin can be achieved.

Drawback:

- Complicated design procedure.

Additional material

There are some videos on Lead and Lag compensators:

- What are Lead and Lag compensators?
<https://www.youtube.com/watch?v=xLhvil5sDcU>
- Designing a Lead compensator with Bode plots
<https://www.youtube.com/watch?v=rH44ttR3G4Q>
- Designing a Lag compensator with Bode plots
<https://www.youtube.com/watch?v=-4bY4W0hvFA>

Remember:

- A Lead compensator is a PD controller with limited gain at high frequencies
- A Lag compensator is a PI controller with limited gain at low frequencies
- A filtered PID controller can be understood as a PI controller combined with a lead compensator.